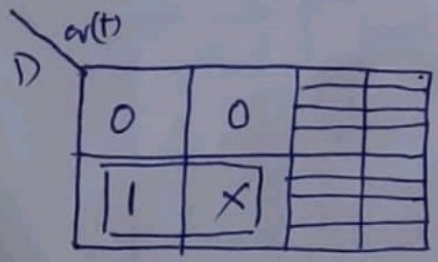
Flip-flop Conversion

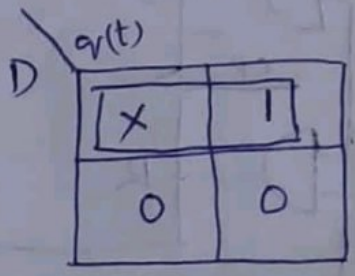
i) SR flip-flop to D flip-flop

Conversion Table

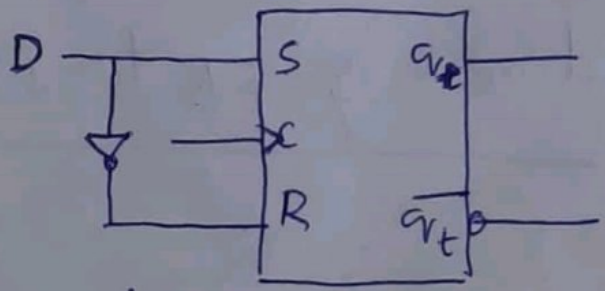
D I/p	outputs		SR I/p	
D	q_t	q_{t+1}	S	R
0	0	0	0	X
1	0	1	1	0
0	1	0	0	1
1	1	1	X	0



$S = D$



$R = \overline{D}$

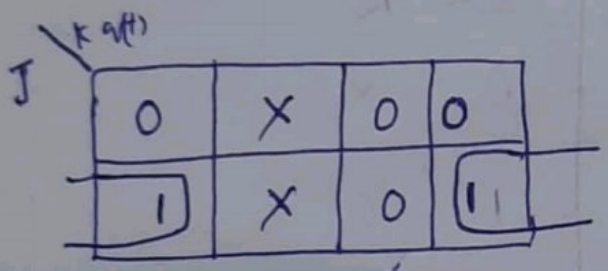


logical diagram of SR to D Conversion FF

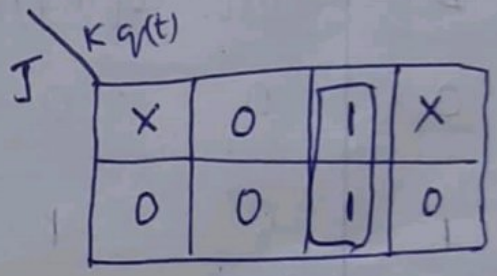
Scanned by CamScanner
Scanned with CamScanner

ii) 14 SR flip-flop JK flip-flop

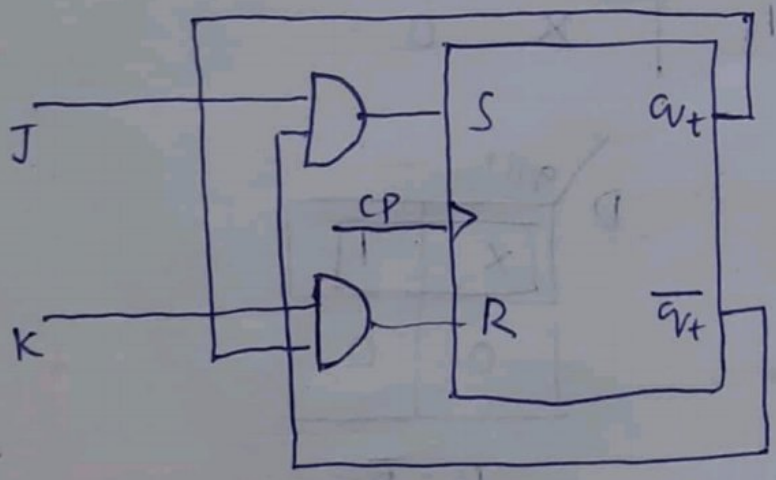
J	K	$q_t(t)$	$q_t(t+1)$	S	R
0	X	0	0	0	X
1	X	0	1	1	0
X	1	1	0	0	1
X	0	1	1	X	0



$S = J\bar{q}_t$



$R = K q_{(t)}$



logical diagram of SR to JK flip-flop

Scanned by CamScanner
Scanned with CamScanner

iii) SR to T flip-flop

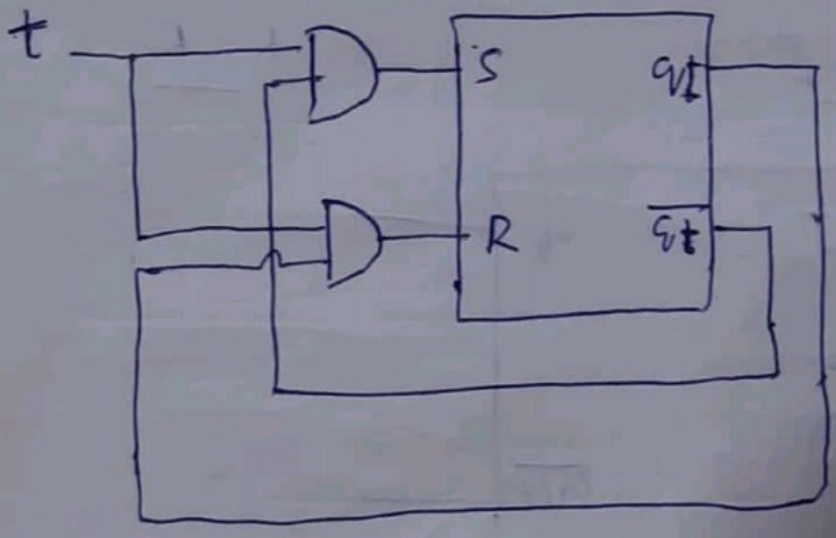
t_{FF}	q_t	q_{t+1}	S	R FF
0	0	0	0	X
1	0	1	1	0
1	1	0	0	1
0	1	1	X	0

t \ q_t	0	1
0	X	0
1	(1)	0

$S = t\,q_t'$

t \ q_t	0	1
0	X	0
1	0	(1)

$R = t\,q_t$

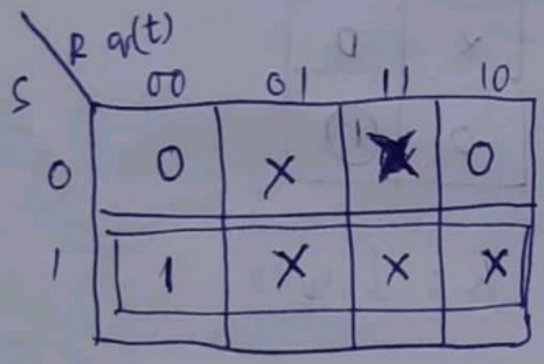


SR to T flip-flop Conversion

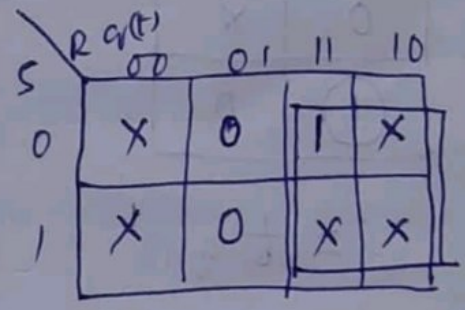
Scanned by CamScanner
Scanned with CamScanner

16(v) JK to SR flip-flop

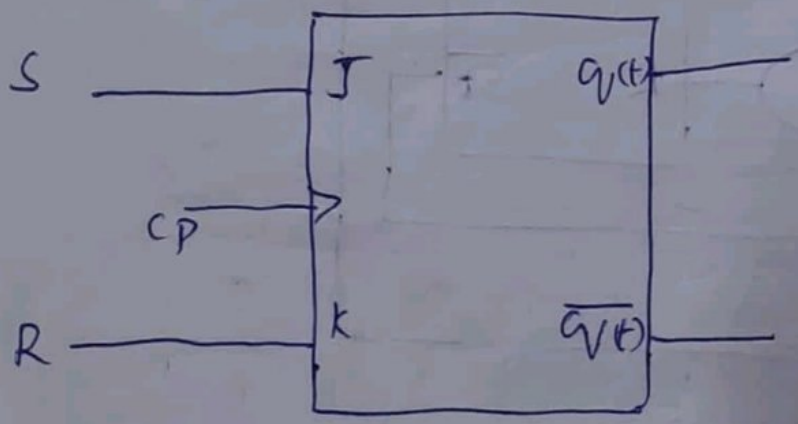
S	R	$q_n(t)$	$q_n(t+1)$	J	K
0	X	0	0	0	X
1	0	0	1	1	X
0	1	1	0	X	1
X	0	1	1	X	0



$J = S$



$K = R$

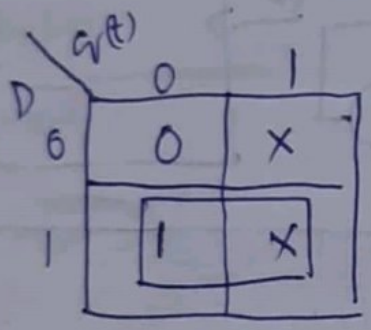


logical diagram of JK to SR

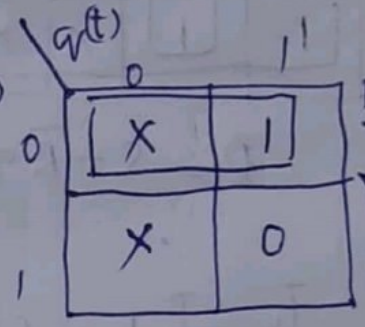
Scanned by CamScanner
Scanned with CamScanner

v) JK to D flip-flop

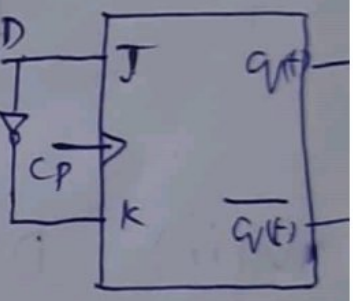
D	$q_n(t)$	$q_n(t+1)$	J	K
0	0	0	0	X
1	0	1	1	X
0	1	0	X	1
1	1	1	X	0



$J = D$

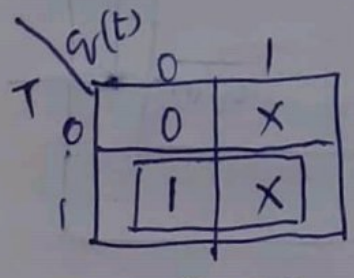


$K = \overline{D}$

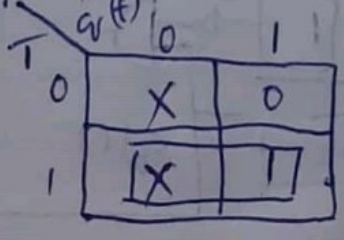


vi) JK to T flip-flop

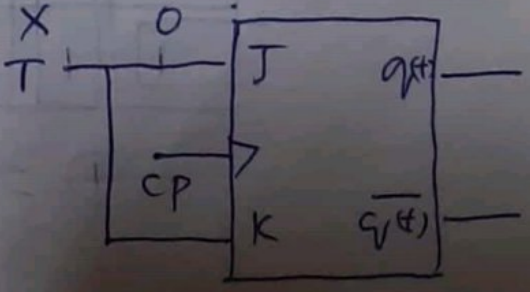
T	$q_n(t)$	$q_n(t+1)$	J	K
0	0	0	0	X
1	0	1	1	X
1	1	0	X	1
0	1	1	X	0



$J = T$



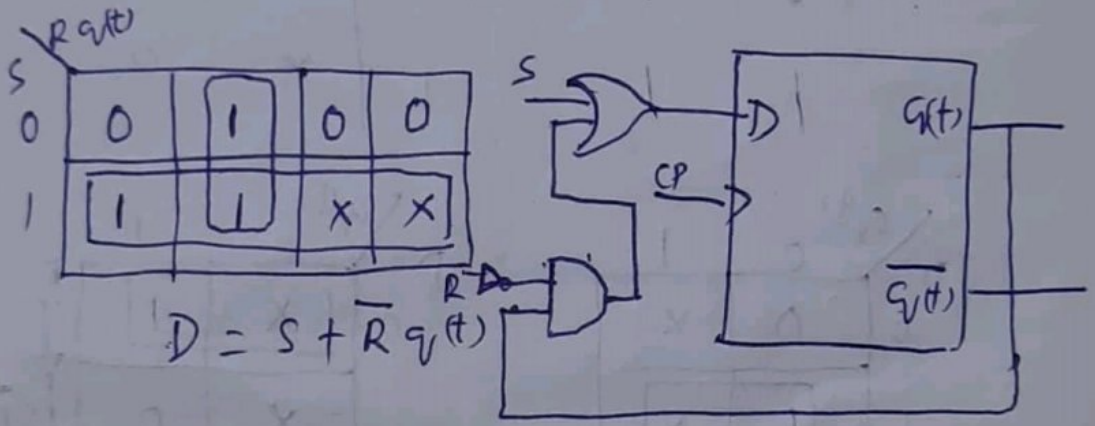
$K = T$



Scanned by CamScanner
Scanned with CamScanner

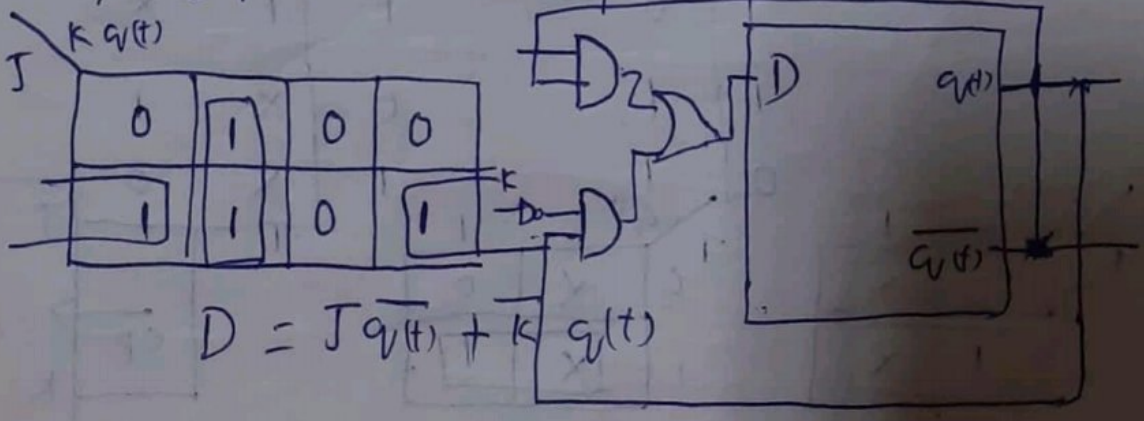
vii) D to SR flip-flop

S	R	$q_v(t)$	$q_v(t+1)$	D
0	X	0	0	0
1	0	0	1	1
0	1	1	0	0
X	0	1	1	1



viii) D to JK flip-flop

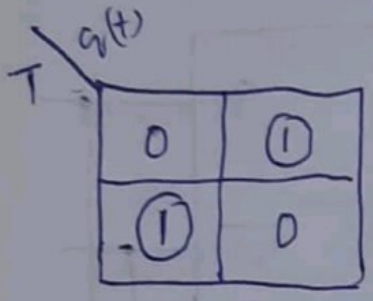
J	K	$q_v(t)$	$q_v(t+1)$	D
0	X	0	0	0
1	X	0	1	1
X	1	1	0	0
X	0	1	1	1



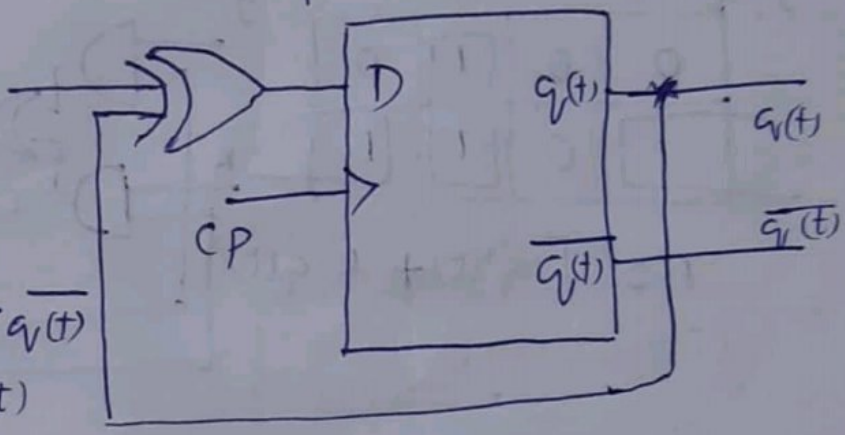
Scanned by CamScanner
Scanned with CamScanner

ix) D flip-flop to T flip-flop

T	$q_n(t)$	$q_n(t+1)$	D
0	0	0	0
1	0	1	1
1	1	0	0
0	1	1	1

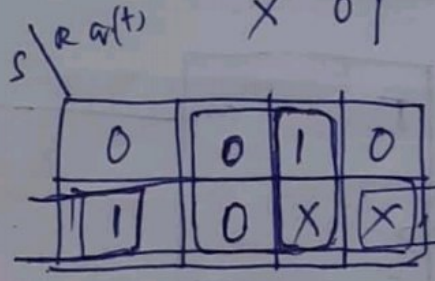


$$D = \bar{T}\, q_n(t) + T\, \overline{q_n(t)}$$
$$= T \oplus q_n(t)$$

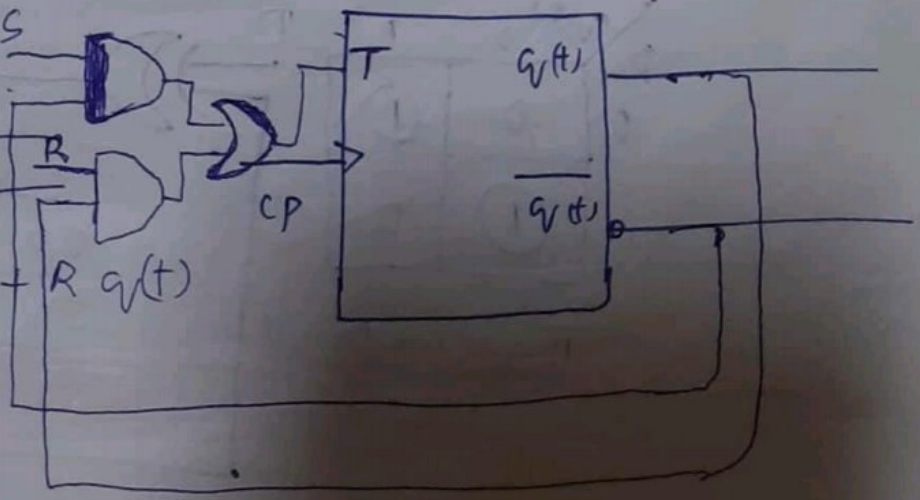


x) To Convert from T to SR flip-flop.

S	R	$q_n(t)$	$q_n(t+1)$	T
0	X	0	0	0
1	0	0	1	1
0	1	1	0	1
X	0	1	1	0



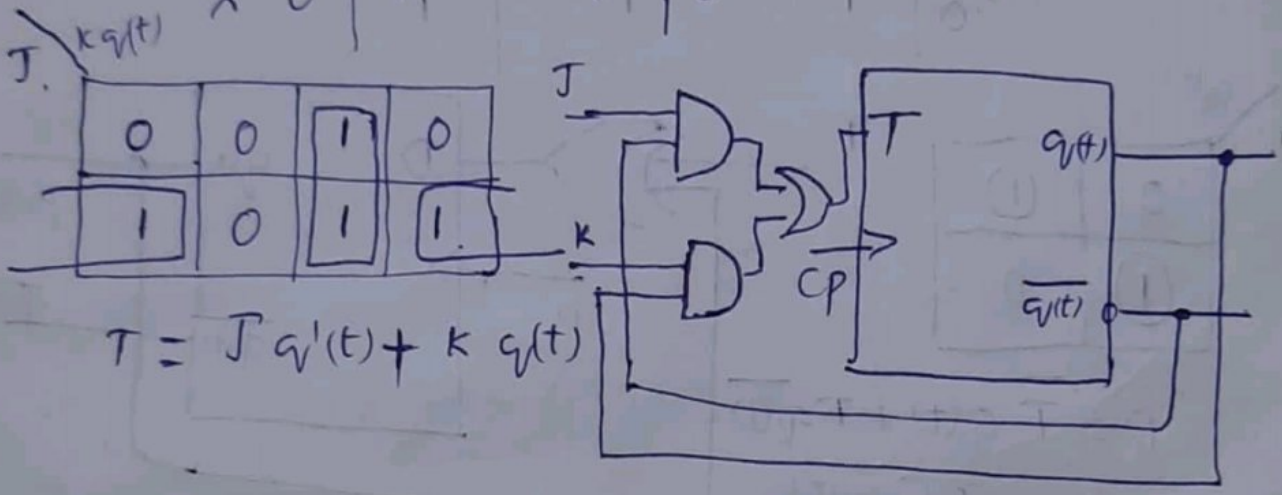
$$T = S\, q'(t) + R\, q(t)$$



Scanned by CamScanner
Scanned with CamScanner

xi) T to JK flip-flop

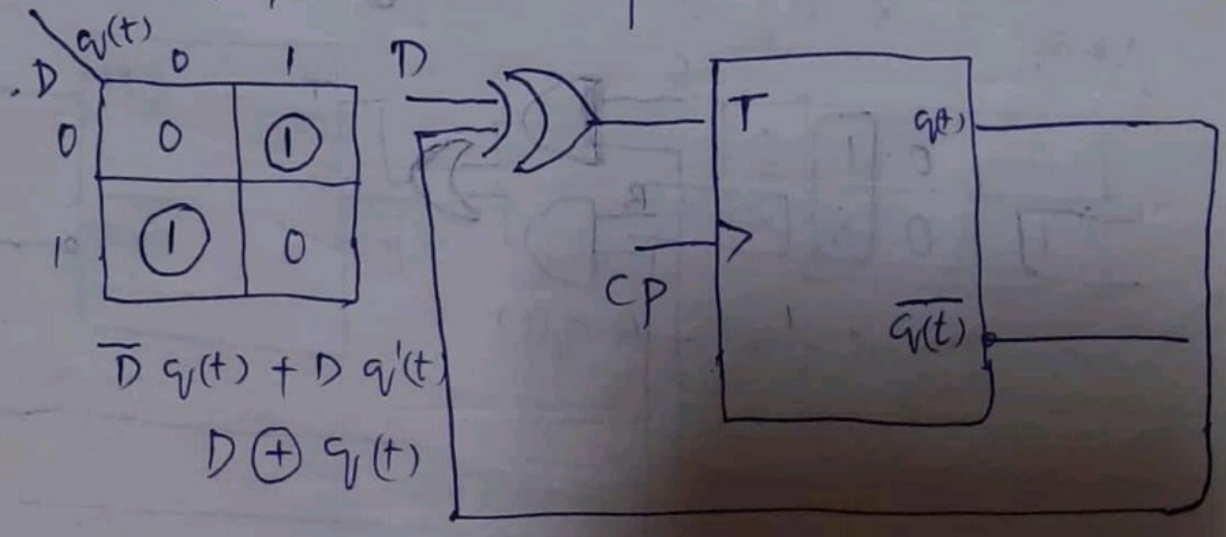
J	K	$q(t)$	$q(t+1)$	T
0	x	0	0	0
1	x	0	1	1
x	1	1	0	1
x	0	1	1	0



$T = J q'(t) + K q(t)$

xii) T to D flip-flop

D	$q(t)$	$q(t+1)$	T
0	0	0	0
1	0	1	1
0	1	0	1
1	1	1	0



$\bar{D}\, q(t) + D\, q'(t)$

$D \oplus q(t)$

Scanned by CamScanner
Scanned with CamScanner

i) A PN flip-flop has four operations, clear to 0, no change, complement, and set to 1, when inputs P & N are 00, 01, 10 & 11, respectively

i) Tabulate the characteristic table

ii) Derive the characteristic equation.

iii) Tabulate the excitation table

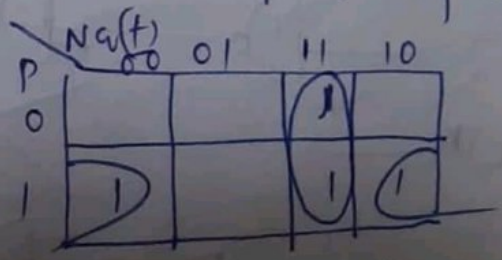
iv) show how the PN flip-flop can be converted to a D flip-flop.

Ans :- i)

P	N	q(t+1)
0	0	0
0	1	q(t)
1	0	q'(t)
1	1	1

ii)

q(t)	P	N	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



$q(t+1) = Pq' + Nq$

Scanned by CamScanner

Scanned with CamScanner

22 iii)

$Q(t)$	$Q(t+1)$	P	N
0	0	0	x
0	1	1	x
1	0	x	0
1	1	x	1

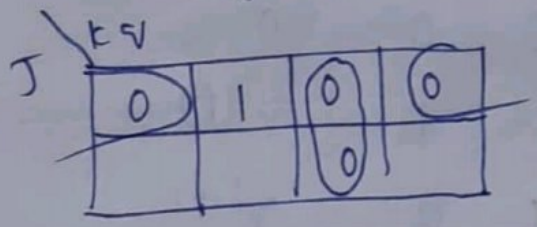
iv) Connect p & N together.

2) ST the characteristic Equation for the complement o/p q a JK FF is

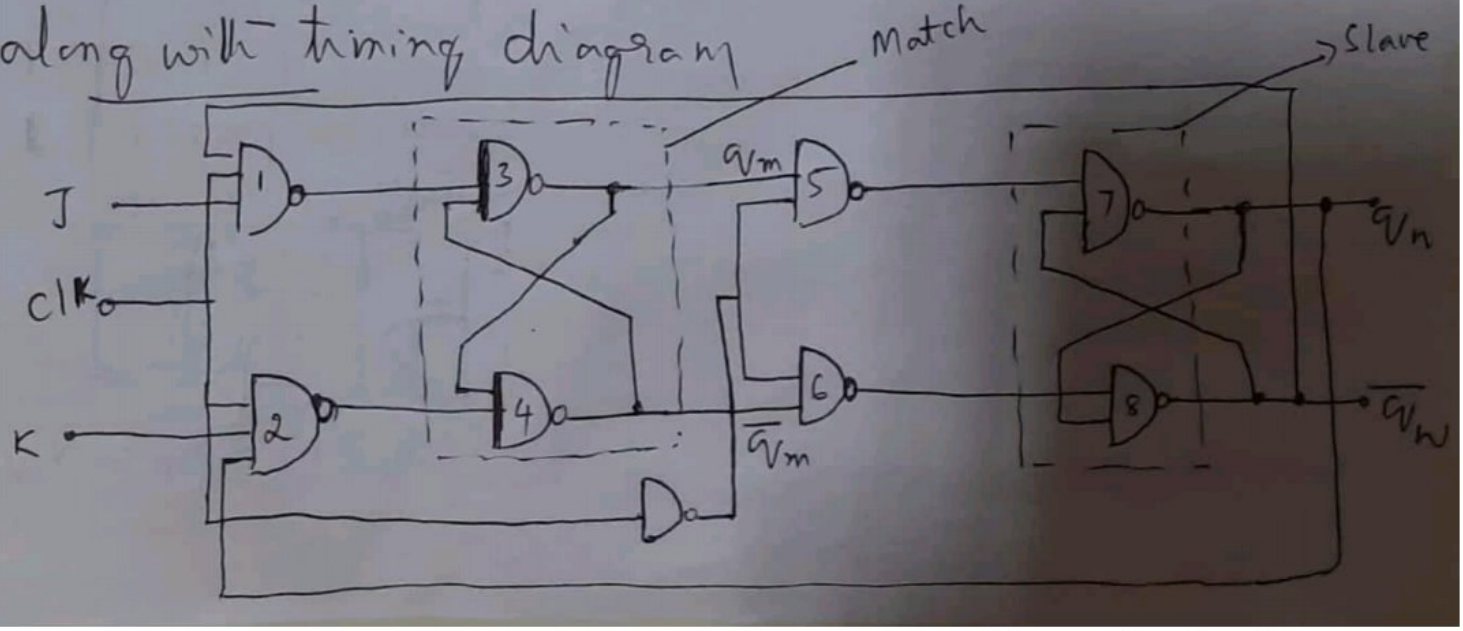
$$Q'(t+1) = J'Q' + KQ$$

$$= (JQ' + K'Q)'$$

$$= (J' + Q)(K + Q') = \bar{J}'Q' + KQ$$



Construction of master-slave flip-flop using JK along with timing diagram

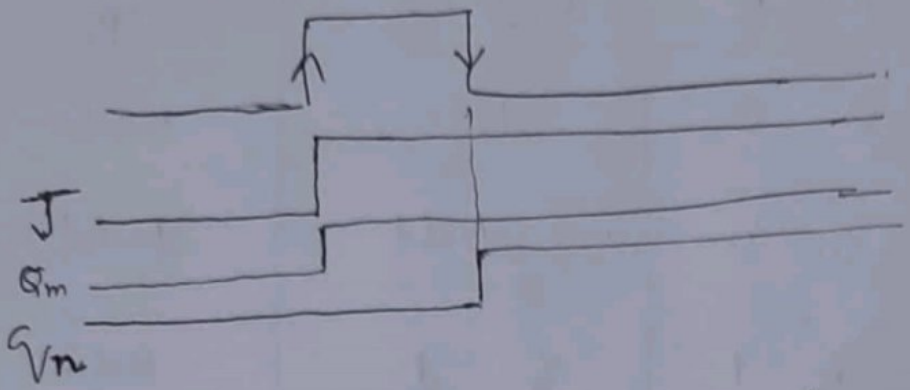


Scanned by CamScanner
Scanned with CamScanner

CP	J	K	Q_m $\overline{Q_m}$	Q_n $\overline{Q_n}$
0 → 1	0	0	Hold	Hold
1 → 0	0	0	Hold	Hold
0 → 1	0	1	0 1	Hold
1 → 0	0	1	Hold	0 1
0 → 1	1	0	1 0	Hold
1 → 0	1	0	Hold	1 0
0 → 1	1	1	Toggle	Hold
1 → 0	1	1	Hold	Toggle

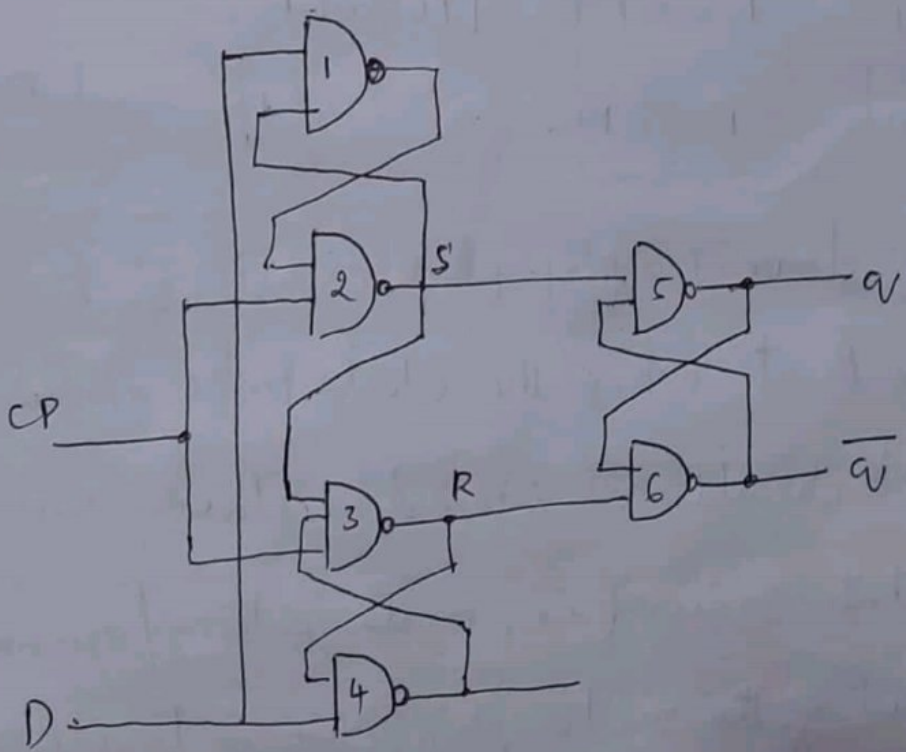
The master-slave JK flip-flop consists of two flip-flops arranged so that when the clock pulse enables the first, or master, it disables the second, or slave. When the clock changes state again (i.e., on its falling/raising edge) the output of the master flip-flop is transferred to the slave flip-flop. Again, toggling is accomplished by the connection of the output with the input gates.

Scanned by CamScanner
Scanned with CamScanner



Timing relationship in a master-slave FF

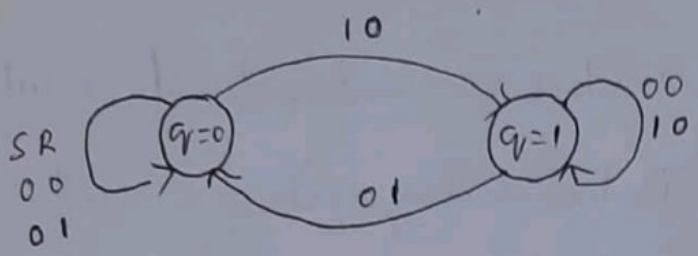
Explain, Edge-triggered D flip-flop



Scanned by CamScanner
Scanned with CamScanner

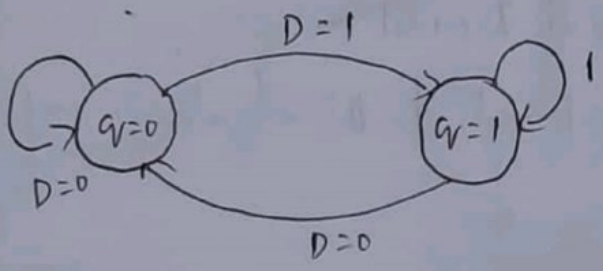
Represent the state transition diagram for all types of FF

q(t)	S	R	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	x
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	x



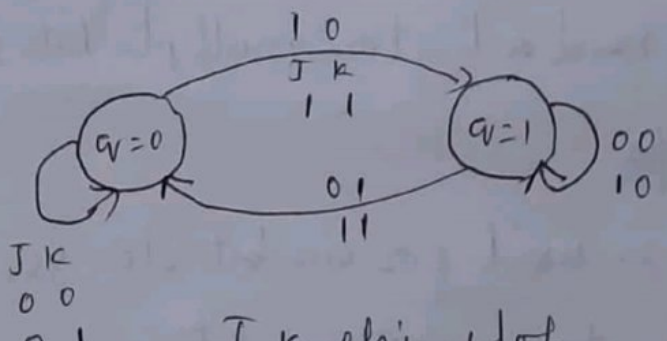
SR flip-flop

q	D	q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1



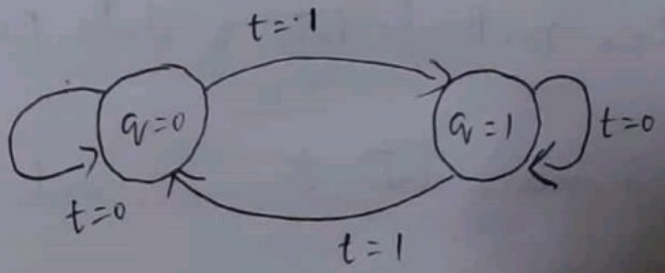
D flip-flop

q	J	K	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



JK flip-flop

q	t	q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0



T flip-flop

Scanned by CamScanner
Scanned with CamScanner

26 Write the differences among a truth table, state table, characteristic table and an excitation table.

Truth table :- describes a combinational circuit

State table :- describes a sequential circuit.

Characteristic table :- describes the operation of a flip-flop.

excitation table :- it gives the values of flip-flop inputs for a given state transition.

The four equations correspond to the algebraic expressions of the 4 tables.

Registers :- Flip flops can store a single bit of binary data, i.e., 1 or 0. But if we need to store multiple bits of data, we need multiple flip-flops

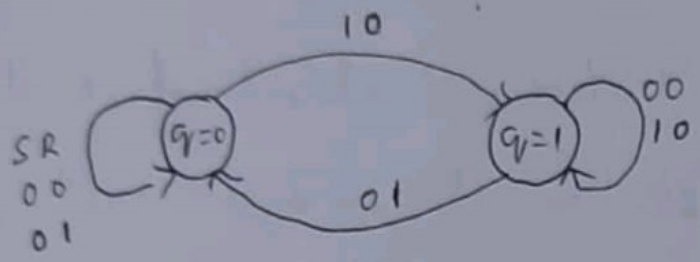
As a single flip-flop is used for one bit storage, n flip-flops are connected in order to store n bits of data.

Flip-flops are use in constructing registers. Registers is a group of flip-flop used to store multiple bits of data. For example, if a computer is to store 16 bits of data, then it needs a set of 16 flip-flops. The inputs and outputs of a register may be serial or parallel based on the requirement.

Scanned by CamScanner

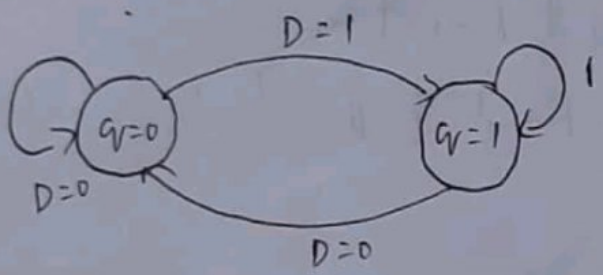
Scanned with CamScanner

Represent the state transition diagram for all types of FF



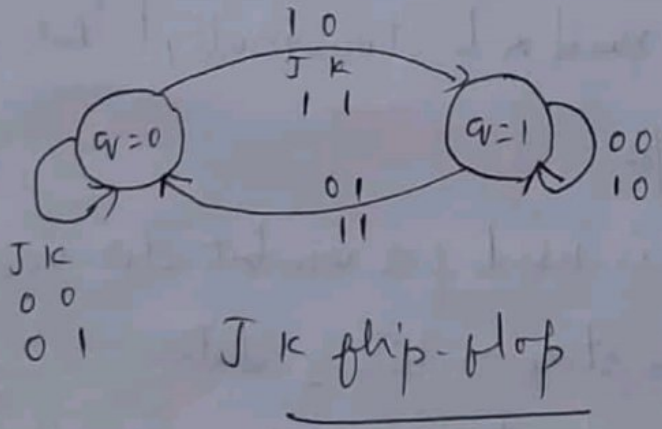
SR flip-flop

q(t)	S	R	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	x
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	x



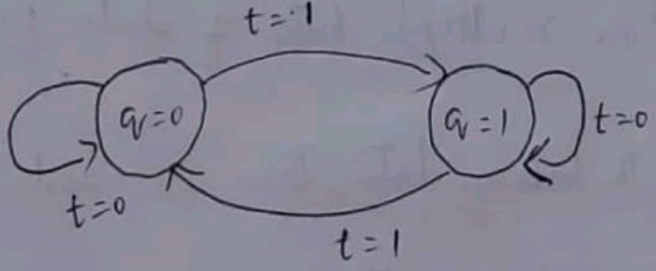
D flip-flop

q	D	q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1



J K flip-flop

q	J	K	q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



T flip-flop

q	t	q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

Scanned by CamScanner
Scanned with CamScanner

When a number of flip-flops are connected in series, to arrange-ment is called a register. The stored information can be transferred within the registers; these are called as shift registers. A shift register is a sequential circuit which stores the data and shifts it towards the output on every clock cycle.

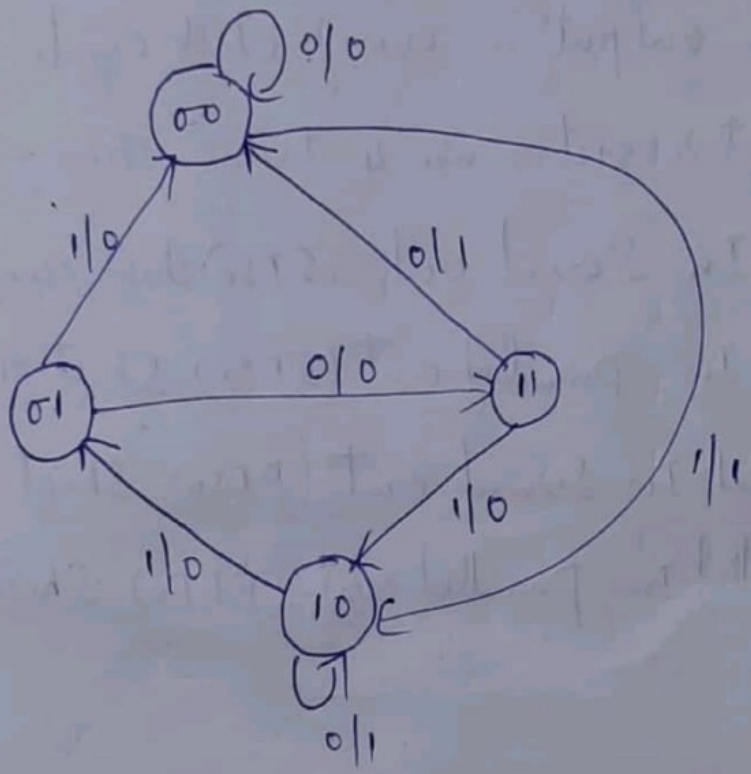
Basically shift register are 4 types. They are

i) Serial In Serial out (SISO) shift register
ii) Serial In parallel out (SIPO) Shift register.
iii) parallel In Serial out (PISO) Shift register
iv) parallel In parallel out (PIPO) Shift register.

Scanned by CamScanner
Scanned with CamScanner

28

1) A Sequential circuit has one input and one output. The state diagram is shown in figure below. Design the sequential circuit with (a) D flip-flop. ii) T flip-flop iii) RS flip-flop. iv) JK flip-flop



State table

P.S	N.S		O/P	
	X=0	X=1	X=0	X=1
AB	$\overline{A}\overline{B}$	AB	Y	Y
00	00	10	0	1
01	11	00	0	0
10	10	01	1	0
11	00	10	1	0

Scanned by CamScanner

Scanned with CamScanner

Design using D flip-flops

A \ BX	00	01	11	10
0	0	(1)	0	(1)
1	(1)	0	(1)	0

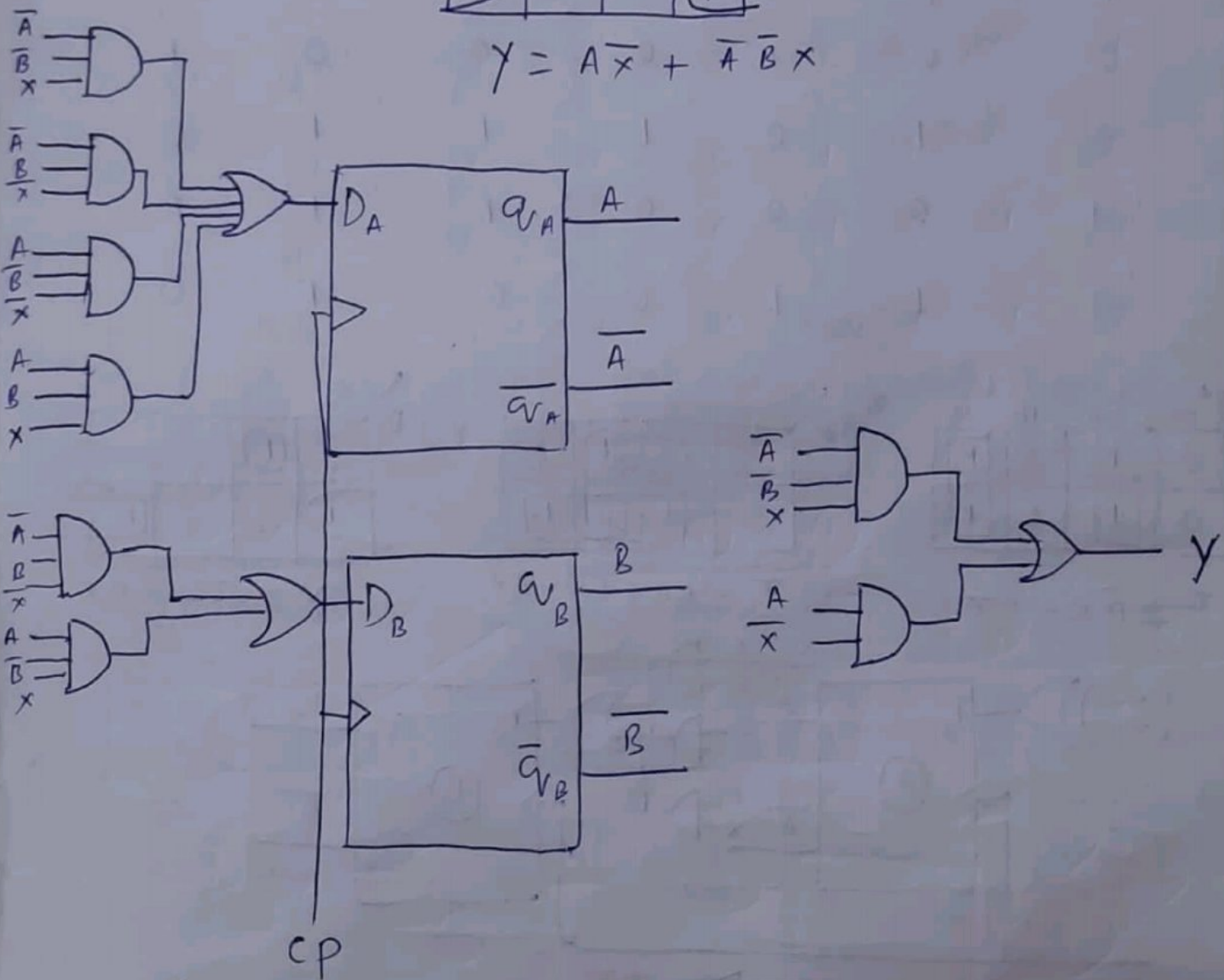
$D_A = \bar{A}\,\bar{B}X + \bar{A}B\bar{X} + A\bar{B}\bar{X} + ABX$

A \ BX	00	01	11	10
0	0	0	0	(1)
1	0	(1)	0	0

$D_B = \bar{A}B\bar{X} + A\bar{B}X$

A \ BX	00	01	11	10
0	0	(1)	0	0
1	(1)	0	0	(1)

$Y = A\bar{X} + \bar{A}\bar{B}X$



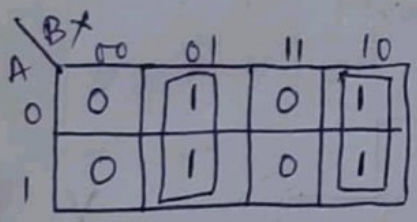
Logic diagram of given sequential circuit using D flip-flop

Scanned by CamScanner

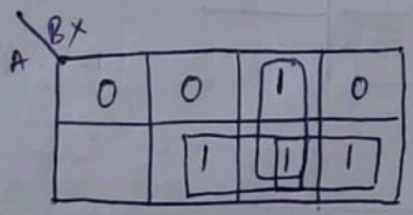
Scanned with CamScanner

ii) Design using T flip-flops

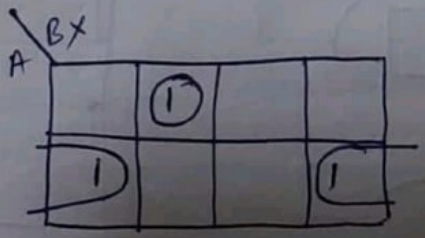
P.S		I/P	N.S		FF I/P		output
A	B	x	A^+	B^+	T_A	T_B	Y
0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	1
0	1	0	1	1	1	0	0
0	1	1	0	0	0	1	0
1	0	0	1	0	0	0	1
1	0	1	0	1	1	1	0
1	1	0	0	0	1	1	1
1	1	1	1	0	0	1	0



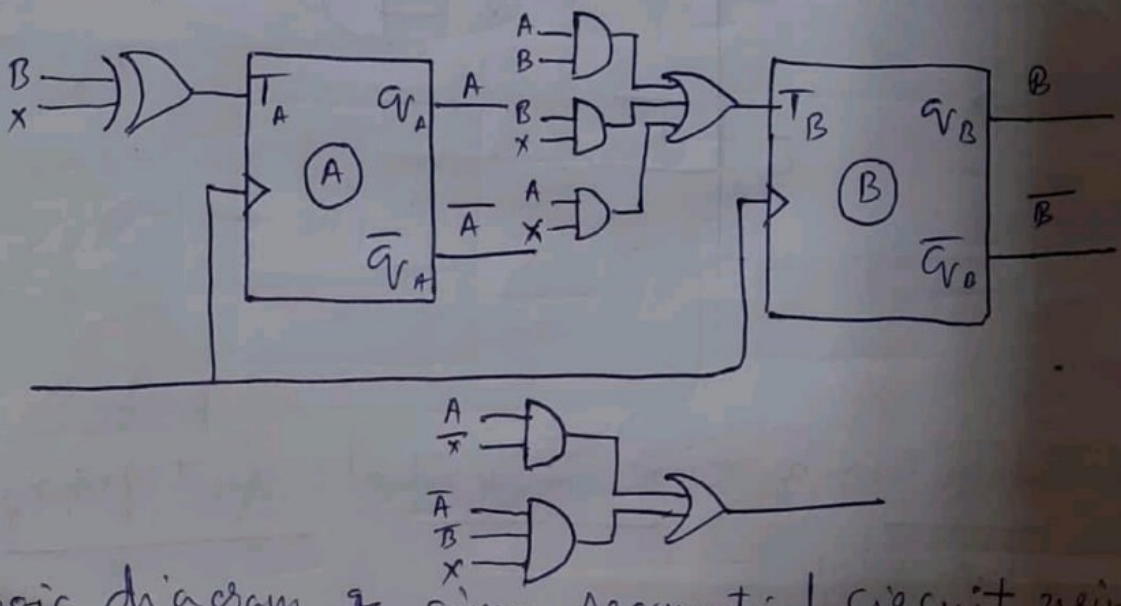
$T_A = \bar{B}x + B\bar{x}$



$T_B = AB + Bx + Ax$



$Y = A\bar{x} + \bar{A}\bar{B}x$

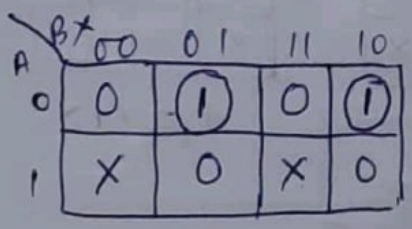


Logic diagram of given sequential circuit using T flip-flop.

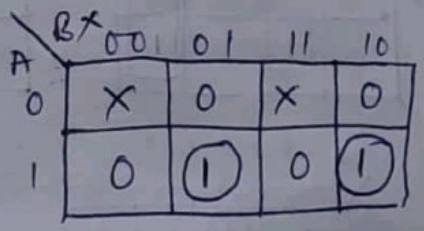
Scanned by CamScanner
Scanned with CamScanner

Design using RS flip-flops

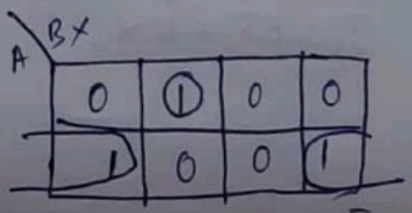
P.S		I/P	N.S		FF I/P				O/P
A	B	X	A^+	B^+	S_A	R_A	S_B	R_B	Y
0	0	0	0	0	0	X	0	X	0
0	0	1	1	0	1	0	0	X	1
0	1	0	1	1	1	0	X	0	0
0	1	1	0	0	0	X	0	1	0
1	0	0	1	0	X	0	0	X	1
1	0	1	0	1	0	1	1	0	0
1	1	0	0	0	0	1	0	1	1
1	1	1	1	0	X	0	0	1	0



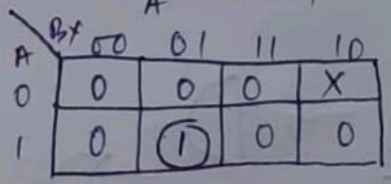
$S_A = \bar{A}B\bar{X} + \bar{A}\bar{B}X$



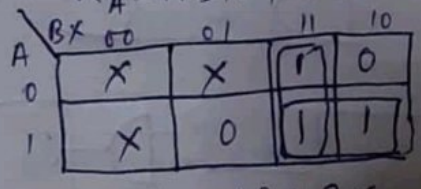
$R_A = A\bar{B}X + AB\bar{X}$



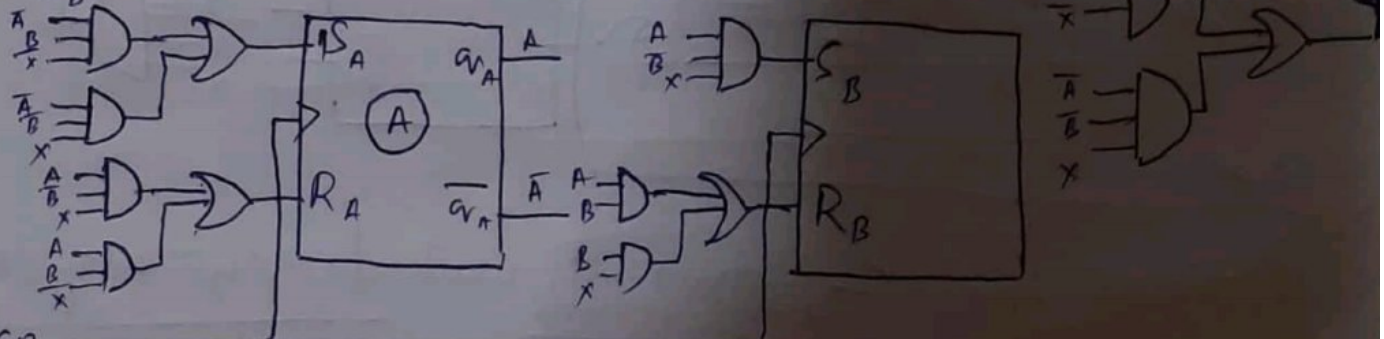
$Y = A\bar{X} + \bar{A}\bar{B}X$



$S_B = A\bar{B}X$



$R_B = AB + BX$

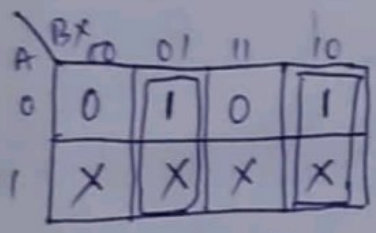


Logic diagram of given sequential circuit using SR flip-flop

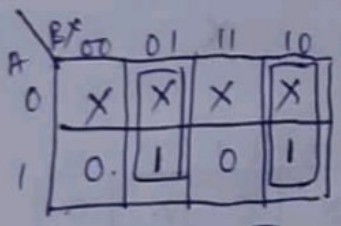
Scanned by CamScanner
Scanned with CamScanner

Design using JK flip-flop

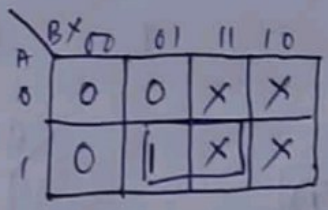
P.S		I/P	N.S		FF I/P				output
A	B	x	A^+	B^+	J_A	K_A	J_B	K_B	Y
0	0	0	0	0	0	x	0	x	0
0	0	1	1	0	1	x	0	x	1
0	1	0	1	1	1	x	x	0	0
0	1	1	0	0	0	x	x	1	0
1	0	0	1	0	x	0	0	x	1
1	0	1	0	1	x	1	1	x	0
1	1	0	0	0	x	1	x	1	1
1	1	1	1	0	x	0	x	1	0



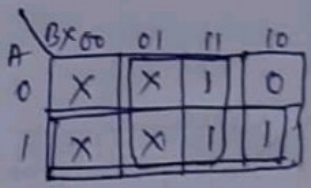
$J_A = B\bar{x} + \bar{B}x$



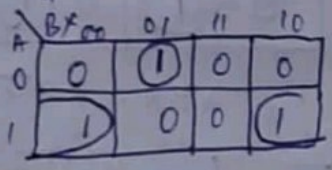
$K_A = B\bar{x} + \bar{B}x$



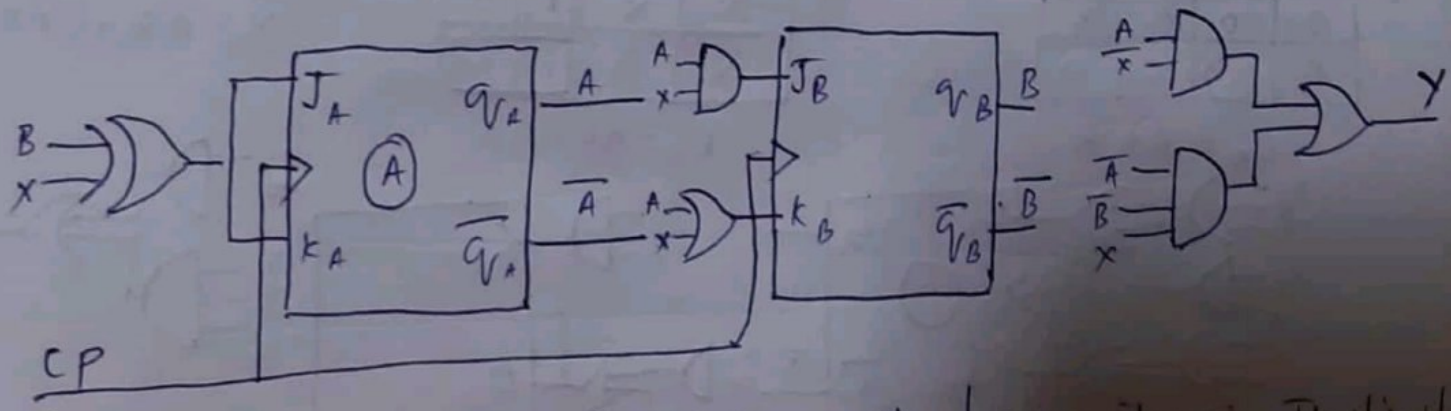
$J_B = Ax$



$K_B = A + x$



$Y = A\bar{x} + \bar{A}\bar{B}x$



logic diagram of given sequential circuit using JK flip-flop

Scanned by CamScanner
Scanned with CamScanner